Linear Algebra and Its Applications

Section 7.S: Supplementary Exercises

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

Contents

7.S Chapter 7 supplementary exercises 3

Exercise 7.S.1: Seventeen true–false statements 3

Exercise 7.S.2: Build a symmetric matrix with prescribed eigenvectors 4

Exercise 7.S.3: Rank counts the nonzero spectral summands 4

Exercise 7.S.4: An orthogonal column/null decomposition for a symmetric matrix 4

Exercise 7.S.5: A nonzero-eigenvalue eigenvector lies in the column space 4

Exercise 7.S.6: Construct the decomposition using an eigenbasis 5

Exercise 7.S.7: Prove the Cholesky characterization 5

Exercise 7.S.8: Convert Cholesky into an LU factorization 5

Exercise 7.S.9: A Gram matrix has the same rank as its original matrix 5

Exercise 7.S.10: Construct a rank-revealing Gram factor 5

Exercise 7.S.11: A polar decomposition, including singular matrices 6

Exercise 7.S.12: The pseudoinverse produces two orthogonal projections 6
Exercise 7.S.13: A consistent system's unique minimum-length solution 6
Exercise 7.S.14: The minimum-length least-squares solution for arbitrary data 6
Exercise 7.S.15: A pseudoinverse and an inconsistent printed right side 7
Exercise 7.S.16: A second minimum-length least-squares computation 7

7.S Chapter 7 supplementary exercises

These exercises connect symmetric spectral decompositions, quadratic-form positivity, Gram matrices, singular value decompositions, and minimum-length solutions. A symmetric matrix is positive definite when $x^TAx > 0$ for every $x \neq 0$, not merely for one vector. A pseudoinverse provides a least-squares solution even when an exact solution does not exist; that distinction is important in the final two printed exercises.

Exercise 7.S.1: Seventeen true–false statements

(a) **True.** If $A = PDP^T$ with P orthogonal and D diagonal, then $A^T = PD^TP^T = A$.

(b) **False.** The 90° rotation $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is orthogonal but is not symmetric.

(c) **True.** $\|Ax\|^2 = x^TA^TAx = x^Tx$ for orthogonal A. Taking nonnegative square roots gives the length equality.

(d) **False.** Principal axes require mutually perpendicular eigenvector directions. For $A = I_2$, every invertible P diagonalizes A, but $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has nonorthogonal columns, so its columns do not give orthogonal principal axes.

(e) **False.** Orthogonal columns need not be unit columns. For $P = 2I$, $P^T = 2I \neq I/2 = P^{-1}$.

(f) **False.** All coefficients of $x_1^2 + x_2^2 + 4x_1x_2$ are positive, but its value at $(1, -1)$ is -2.

(g) **False.** $x_1^2 - x_2^2$ is positive at e_1 and negative at e_2, so positivity at one input does not give definiteness.

(h) **True.** The symmetric coefficient matrix admits an orthonormal eigenbasis, and its orthogonal change of variables removes all mixed terms by the Principal Axes Theorem.

(i) **False.** For $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, the largest diagonal entry is 1, but the unit vector $(1, 1)^T/\sqrt{2}$ gives $x^TAx = 2$, the largest eigenvalue.

(j) **False without a constraint.** A positive definite form is unbounded above: $Q(tx) = t^2Q(x) \to \infty$ for fixed $x \neq 0$. The greatest eigenvalue is the maximum only when $\|x\| = 1$ is imposed.

(k) **False.** If P is orthogonal, nonzero u gives $x = Pu \neq 0$, so $(Pu)^TA(Pu) > 0$. An invertible variable change preserves positive definiteness rather than turning it negative.

(l) **False.** An indefinite form has a symmetric coefficient matrix with both positive and negative eigenvalues. “Not definite eigenvalues” is not a mathematical alternative definition; semidefinite forms are not indefinite merely because an eigenvalue is zero.

(m) **True.** Substitution gives $(Pu)^TA(Pu) = u^TP^TAPu = u^T(P^{-1}AP)u$.

(n) **False.** Orthonormal columns are required for UU^T to be the projection formula. For $U = 2e_1$ and $x = e_1$, $UU^Tx = 4e_1$, whereas the projection onto its column line is e_1.

(o) **True.** An SVD gives $\|Bx\|^2 = \sum_i \sigma_i^2 y_i^2 \le \sigma_1^2 \sum_i y_i^2 = \sigma_1^2$ for $y = V^T x$ and $\|x\| = 1$.

(p) **True as an existence statement.** From an ordinary SVD $B = U\Sigma V^T$, take $P = U$ and $Q = V^T$. The transpose of an orthogonal matrix is orthogonal, so $B = P\Sigma Q$ has exactly the asserted form. This does not mean one can omit a transpose while keeping the same right singular-vector matrix V; the new symbol Q denotes its transpose.

(q) **False.** The singular values of $A^T A$ are the squares of those of A. For $A = 2I$, they are 4 versus 2, so the lists differ.

Exercise 7.S.2: Build a symmetric matrix with prescribed eigenvectors

(a) Each outer product satisfies $(u_i u_i^T)^T = u_i u_i^T$. A real linear combination of symmetric matrices is symmetric, so $A^T = A$.

(b) For each j, orthonormality gives

$$Au_j = \sum_i \lambda_i u_i (u_i^T u_j) = \lambda_j u_j.$$

Every u_j is nonzero, so it is an eigenvector with eigenvalue λ_j. Because these n vectors are a basis, the list contains all eigenvalues, including their multiplicities.

Exercise 7.S.3: Rank counts the nonzero spectral summands

Write $A = PDP^T$ with orthogonal P. Multiplication by invertible matrices preserves rank, so $\operatorname{rank} A = \operatorname{rank} D$, the number of nonzero diagonal eigenvalues. Thus exactly r terms in $A = \sum_i \lambda_i u_i u_i^T$ are nonzero. For each such term, every column is a multiple of u_i, and at least one column is nonzero because $u_i \neq 0$ and $\lambda_i \neq 0$. Each nonzero term therefore has rank one. Discarding zero terms represents A as a sum of r rank-one matrices, as claimed. When $r = 0$, this is the empty sum representing zero.

Exercise 7.S.4: An orthogonal column/null decomposition for a symmetric matrix

(a) A vector z is perpendicular to every column of A precisely when $A^T z = 0$. Since $A^T = A$, this says $z \in \operatorname{Nul} A$. Therefore $(\operatorname{Col} A)^\perp = \operatorname{Nul} A$.

(b) Orthogonally decompose y relative to $W = \operatorname{Col} A$: $y = \widehat{y} + z$, where $\widehat{y} \in \operatorname{Col} A$ and $z \in (\operatorname{Col} A)^\perp = \operatorname{Nul} A$. Such a decomposition exists by the Orthogonal Decomposition Theorem and is unique.

Exercise 7.S.5: A nonzero-eigenvalue eigenvector lies in the column space

If $Av = \lambda v$ with $\lambda \neq 0$, then $v = A(v/\lambda)$. This explicitly writes v as an output of A, so $v \in \operatorname{Col} A$. The nonzero-eigenvalue condition permits division; a zero-eigenvalue eigenvector need not belong to the column space.

Exercise 7.S.6: Construct the decomposition using an eigenbasis

Choose an orthonormal eigenbasis u_i and write $y = \sum_i c_i u_i$. Separate the sum into

$$\widehat{y} = \sum_{\lambda_i \neq 0} c_i u_i, \qquad z = \sum_{\lambda_i = 0} c_i u_i.$$

Every summand of $\widehat{y}$ belongs to $\operatorname{Col} A$ by Exercise 5, so $\widehat{y} \in \operatorname{Col} A$. Every summand of z is annihilated by A, so $z \in \operatorname{Nul} A$. Their sum is y, giving the requested second proof. The two sums are perpendicular because they use disjoint parts of an orthonormal basis.

Exercise 7.S.7: Prove the Cholesky characterization

A Cholesky factorization implies positive definiteness.

If $A = R^T R$ with invertible real R, it is symmetric and $x^T A x = \|Rx\|^2 > 0$ whenever $x \neq 0$. Thus A is positive definite.

A positive definite matrix supplies such a factorization.

Exercise 7.2.26 gives an invertible positive definite square root B with $A = B^T B$. Take a QR factorization $B = QR$, choosing the signs of columns of Q and rows of R so the nonzero diagonal entries of R are positive. Here Q is square orthogonal and R is upper triangular. Then $A = B^T B = R^T Q^T Q R = R^T R$. The full rank of B makes R invertible, and the sign choices give the required positive diagonal. This proves both directions of the equivalence.

Exercise 7.S.8: Convert Cholesky into an LU factorization

From Exercise 7, $A = R^T R$ with upper triangular R and positive r_{ii}. Let $D = \operatorname{diag}(r_{11}, \ldots, r_{nn})$. Define $L = R^T D^{-1}$, which is lower triangular with diagonal entries one, and $U = DR$, which is upper triangular with diagonal entries $r_{ii}^2 > 0$. Then $LU = R^T D^{-1} DR = R^T R = A$, as required. The parenthetical converse in the text is understood for symmetric matrices and the usual unit-lower-triangular LU convention. Without symmetry, an arbitrary upper triangular matrix with positive diagonal need not be a positive definite *symmetric* matrix.

Exercise 7.S.9: A Gram matrix has the same rank as its original matrix

$G = A^T A$ is symmetric and $x^T G x = \|Ax\|^2 \geq 0$, so it is positive semidefinite. Moreover, $Gx = 0$ implies $0 = x^T G x = \|Ax\|^2$, hence $Ax = 0$. Conversely $Ax = 0$ immediately implies $Gx = A^T 0 = 0$. Thus $\operatorname{Nul} G = \operatorname{Nul} A$. Both matrices have the same number n of columns, so rank–nullity gives $\operatorname{rank} G = n - \dim \operatorname{Nul} G = n - \dim \operatorname{Nul} A = \operatorname{rank} A$. This also proves why forming a Gram matrix does not create new independent column directions in exact arithmetic.

Exercise 7.S.10: Construct a rank-revealing Gram factor

A symmetric positive semidefinite G has r positive eigenvalues $\lambda_1, \ldots, \lambda_r$ and the rest zero. Let U_r contain corresponding orthonormal eigenvectors, and $D_r = \operatorname{diag}(\lambda_1, \ldots, \lambda_r)$. Then $G = U_r D_r U_r^T$.

Choose the $r \times n$ matrix

$$\boxed{A = D_r^{1/2} U_r^T}.$$

Its Gram matrix is $A^T A = U_r D_r U_r^T = G$. Equivalently the hinted factor $B = U_r D_r^{1/2}$ satisfies $G = BB^T$ and $A = B^T$. When $r = 0$, $G = 0$ and the $0 \times n$ empty matrix has zero Gram matrix under the empty-sum convention.

Exercise 7.S.11: A polar decomposition, including singular matrices

Take a square SVD $A = U\Sigma V^T$, with U, V orthogonal and Σ diagonal with nonnegative entries. Set

$$P = U\Sigma U^T, \qquad Q = UV^T.$$

Then P is symmetric positive semidefinite by its orthogonal diagonalization, and $\operatorname{rank} P = \operatorname{rank} \Sigma = \operatorname{rank} A$. Also $Q^T Q = VU^T UV^T = I$, so Q is orthogonal. Finally $PQ = U\Sigma U^T UV^T = U\Sigma V^T = A$. No inverse singular values were used, so the construction works for singular A as well. The orthogonal factor need not be unique in that case.

Exercise 7.S.12: The pseudoinverse produces two orthogonal projections

Use the reduced SVD $A = U_r D V_r^T$ and $A^+ = V_r D^{-1} U_r^T$, where the r singular values in D are positive.

(a) Since $V_r^T V_r = I_r$, $AA^+ = U_r D D^{-1} U_r^T = U_r U_r^T$. The columns of U_r are an orthonormal basis of $\operatorname{Col} A$; therefore $AA^+ y$ is the orthogonal projection of y onto that space.

(b) Similarly $A^+ A = V_r V_r^T$, the projection onto $\operatorname{Col} V_r = \operatorname{Row} A$.

(c) Multiplying the formulas and using orthonormality gives

$$AA^+ A = U_r U_r^T U_r D V_r^T = A, \qquad A^+ AA^+ = V_r V_r^T V_r D^{-1} U_r^T = A^+.$$

Thus both requested triple-product identities hold.

Exercise 7.S.13: A consistent system's unique minimum-length solution

(a) Since $b = Ax$ for some x by consistency, $x^+ = A^+ b = A^+ Ax$ is the projection of x onto $\operatorname{Row} A$ by Exercise 12(b). Hence $x^+ \in \operatorname{Row} A$.

(b) Since $b \in \operatorname{Col} A$, its projection there is itself, so $Ax^+ = AA^+ b = b$ by Exercise 12(a). Thus x^+ is indeed a solution.

(c) For any other solution u, $A(u - x^+) = b - b = 0$. Thus $u - x^+ \in \operatorname{Nul} A$, perpendicular to $x^+ \in \operatorname{Row} A = (\operatorname{Nul} A)^\perp$. Pythagoras yields $\|u\|^2 = \|x^+\|^2 + \|u - x^+\|^2 \geq \|x^+\|^2$. Equality holds only when $u = x^+$. Taking square roots proves the minimum length and its uniqueness.

Exercise 7.S.14: The minimum-length least-squares solution for arbitrary data

Let $\widehat{b} = AA^+ b$, the projection of b onto $\operatorname{Col} A$. A vector x is a least-squares solution of $Ax = b$ exactly when $Ax = \widehat{b}$, since $\widehat{b}$ is the unique closest output. This new equation is consistent. By Exercise

13 its unique minimum-length solution is $A^+\widehat{b}$. The triple-product identity from Exercise 12 gives $A^+\widehat{b} = A^+AA^+b = A^+b$. Thus $\boxed{A^+b}$ is the unique minimum-length least-squares solution, even when the original equation itself has no solution.

Exercise 7.S.15: A pseudoinverse and an inconsistent printed right side

Check consistency before claiming to solve the original system.

Rows 3 and 4 of the printed A are identical, but the prescribed $b = (6, -1, -4, 6)^T$ has unequal entries 3 and 4. Hence $Ax = b$ has *no exact solution*. The requested pseudoinverse calculation is nevertheless well defined: it returns the minimum-length least-squares solution from Exercise 14, not an exact solution from Exercise 13. We retain the printed data and make that distinction explicit.

Compute from an SVD, then verify the exact rational matrix.

An executed numerical SVD has nonzero singular values approximately 9.84443449, 2.62465501, and 1.09466692. Its fourth singular value is at roundoff level, representing an exact zero. In NumPy the construction can be reproduced by

```
U,d,Vt = np.linalg.svd(A,full_matrices=False)
r = np.count_nonzero(d > 1e-12)
Ap = (Vt[:r,:].T/d[:r])@U[:,:r].T
```

The exact rational result, verified by the four pseudoinverse identities, is

$$A^+ = \frac{1}{40}\begin{bmatrix} -2 & -14 & 13 & 13 \\ -2 & -14 & 13 & 13 \\ -2 & 6 & -7 & -7 \\ 2 & -6 & 7 & 7 \\ 4 & -12 & -6 & -6 \end{bmatrix}.$$

Here the verification is $AA^+A = A$, $A^+AA^+ = A^+$, and symmetry of AA^+ and A^+A; all products were checked with exact rational arithmetic. Multiplying by the printed right side gives

$$\widehat{x} = A^+b = (7/10, 7/10, -4/5, 4/5, 3/5)^T, \qquad A\widehat{x} = (6, -1, 1, 1)^T.$$

The residual is $(0, 0, -5, 5)^T$ and is perpendicular to every column of A, since its two nonzero components multiply identical rows with opposite weights. Thus $A^T(b - A\widehat{x}) = 0$, independently checking the least-squares claim.

Check row-space membership and the strict norm inequality.

An explicit witness is $\widehat{x} = A^T(7/100, -91/100, 129/100, 0)^T$, so $\widehat{x} \in \operatorname{Row} A$. Choose $u = (-1, 1, 0, 0, 0)^T$. Its first two columns cancel, so $Au = 0$; also $\widehat{x} \cdot u = 0$. Finally $\|\widehat{x}\|^2 = 131/50$ and $\|\widehat{x} + u\|^2 = 231/50$, proving $\|\widehat{x}\| < \|\widehat{x} + u\|$.

Exercise 7.S.16: A second minimum-length least-squares computation

Again distinguish least squares from exact consistency.

Here row 4 is three times row 3, whereas the prescribed right side has $6 \neq 3(-4)$. Thus the printed system is also inconsistent. Its pseudoinverse solution must be interpreted via Exercise 14.

Construct and verify the pseudoinverse.

The executed SVD's nonzero singular values are approximately $12.9536237, 1.44552724, .337763285$. Using the SVD formula in Exercise 15 and verifying exact products gives

$$A^+ = \frac{1}{20}\begin{bmatrix} 10 & 0 & -1 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 40 & 10 & 30 \\ 10 & -20 & -7 & -21 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

All four pseudoinverse identities hold exactly. The resulting input and fitted output are

$$\widehat{x} = (23/10, 0, 5, -9/10, 0)^T, \qquad A\widehat{x} = (6, -1, 7/5, 21/5)^T.$$

The residual is $(0, 0, -27/5, 9/5)^T$. Its row-weight combination vanishes because row 4 equals three times row 3: $(-27/5)\,\mathrm{row}_3\, A + (9/5)\,\mathrm{row}_4\, A = 0$. Hence the normal equations hold and this is a least-squares solution.

Verify the two geometric checks.

The identity $\widehat{x} = A^T(7/10, 109/10, 27, 0)^T$ places it in the row space. Take the nonzero null vector $u = e_2 = (0, 1, 0, 0, 0)^T$, since column 2 is zero. It is perpendicular to $\widehat{x}$, whose second coordinate is zero. The squared norms are $\|\widehat{x}\|^2 = 311/10$ and $\|\widehat{x} + u\|^2 = 321/10$, giving the requested strict inequality. These checks establish the minimum-length property without falsely claiming that the inconsistent original equation has been solved exactly.